Euclid, Book I, Proposition I.1

To Construct an Equilateral Triangle on a Given Finite Straight Line

CGJteamLab

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Euclid's first proposition in Book I is usually presented as the straightforward construction obtained from the intersection of two circles. A formal reconstruction, however, reveals two logically independent problems: the existence of a common point of the corresponding circles and the proof that the resulting point does not lie on the line determined by the given base.

The proof of non-collinearity relies on the classical idea that a proper part of a segment cannot be congruent to the entire segment. This argument is well known from modern axiomatic reconstructions of geometry, including the work of Michael Beeson and his collaborators.

The purpose of this note is not to present a new geometric proof. Rather, it is to demonstrate that, once Book Zero has been developed, this argument becomes a direct application of the previously established theory. Proposition I.1 therefore serves as the first experimental test of Book Zero as an intermediate language for synthetic proofs.

1 The Classical Configuration and Construction

Given a nondegenerate segment AB, Euclid constructs an equilateral triangle on it.

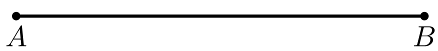

Figure 1: The data of Proposition I.1: a nondegenerate segment AB.

Draw the circle centered at A through B, and the circle centered at B through A. Choose an intersection point C, and join C to A and B.

The metric conclusion is immediate from the definitions of the two circles:

$$AC \cong AB, \qquad BC \cong AB.$$

Therefore the three sides are equal and ABC is equilateral.

Diagrammatic audit. The four successive classical snapshots have been reduced to the initial segment and the completed two-circle construction. The intermediate states merely add the first circle and then the second; they are not distinct geometric cases.

Wilkins draws attention to the genuinely nontrivial point hidden by the familiar picture: Euclid tacitly assumes that the two circles intersect. Thus the diagram should not be read as a proof of intersection. The existence of C requires an additional circle-intersection or continuity principle beyond the mere permission to draw the two circles.

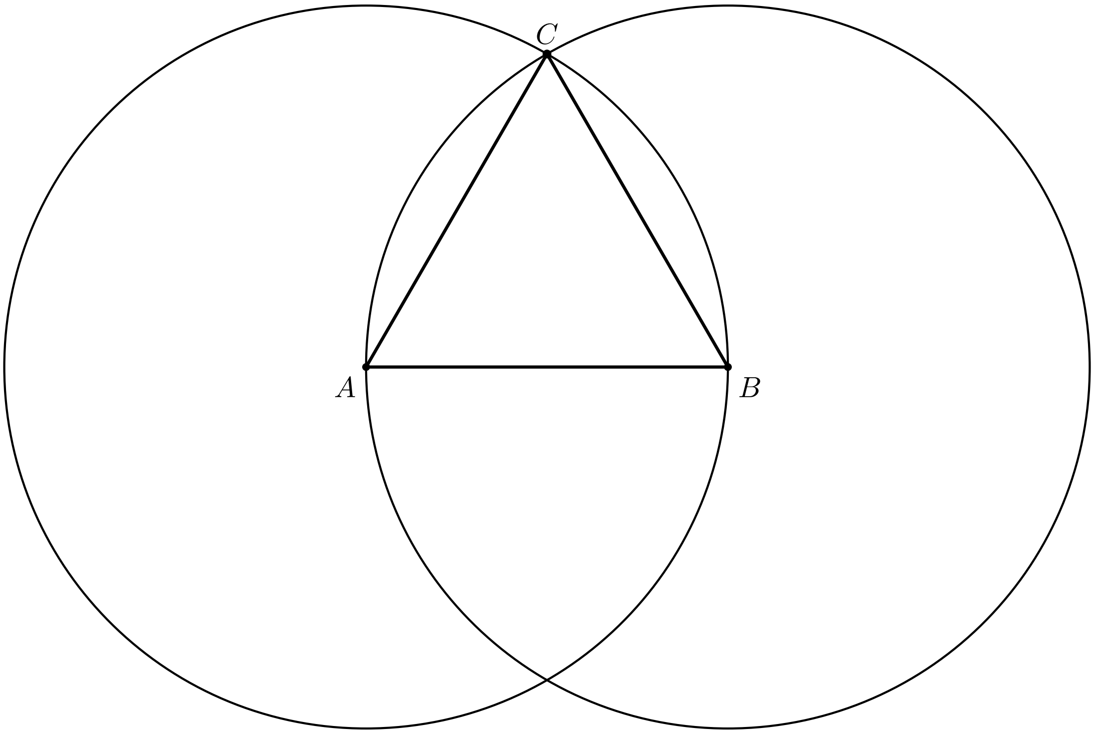

Figure 2: The complete classical construction. Since C lies on both circles, $AC = AB$ and $BC = AB$; hence $AC = BC$ by Common Notion 1.

2 Separating the Two Problems

The reconstruction reveals two logically independent stages.

1. **Existence stage.** One must obtain a point C such that

$$AC \cong AB, \qquad BC \cong AB.$$

   In the current version of the project, this stage is provided by the temporary principle `hilbert_equidistant_point_exists`. Ultimately, it is intended to be derived from an appropriate theory of continuity in Hilbert geometry.

2. **Synthetic stage.** From the congruences above one must derive

$$\neg\,\mathrm{Collinear}(A, B, C).$$

   This stage no longer requires any theory of circles.

3 Theorem

Theorem. Let $A \neq B$. Assume that there exists a point C such that

$$AC \cong AB, \qquad BC \cong AB.$$

Then the points A, B, C are not collinear.

4 Proof

The formal proof separates existence of an equidistant point from proof that the point is genuinely off the base line.

First `hilbert_equidistant_point_exists` supplies a point C with

$$AC \cong AB, \qquad BC \cong AB.$$

The proof then excludes the degenerate possibilities $C = A$ and $C = B$ using the null-segment theorem.

It remains to prove noncollinearity. Assume that A, B, C are collinear. Since the three points are pairwise distinct, betweenness trichotomy gives exactly three possibilities:

$$A - B - C, \qquad B - A - C, \qquad A - C - B.$$

In every case one of the congruent segments would be a proper part of the other. Book Zero #45, `bookZero_45_partNotEqualWhole`, rules this out. Hence A, B, C are not collinear.

This is a significant foundational difference from Euclid. The formal proof does not construct C as an intersection of two circles; existence is provided by the Hilbert interface, and Book Zero order theory is then used to prove that the witness cannot be collinear with A, B.

□

5 Proof Architecture of the Reconstruction

```
Euclid
------
segment AB
   |
circle center A through B
+ circle center B through A
   |
intersection C
   |
AC = AB and BC = AB
   |
Common Notion 1
   |
equilateral ABC


Hilbert / Lean
--------------
A != B
   |
hilbert_equidistant_point_exists
   |
C with AC ~= AB and BC ~= AB
   |
exclude C=A and C=B
```

```
   |
assume A,B,C collinear
   |
betweenness trichotomy
   |
bookZero_45_partNotEqualWhole
in each of the three cases
   |
not collinear A B C
   |
euclid_proposition_1
```

6 Significance of the Reconstruction

The geometric result itself is not new. Constructing an equilateral triangle on a given segment is one of the classical theorems of elementary geometry.

Likewise, the idea underlying the proof of non-collinearity is not new. Modern axiomatic reconstructions, in particular the work of Michael Beeson and his collaborators, employ an analogous argument based on the theorem that a proper part of a segment cannot be congruent to the entire segment.

The novelty of the present reconstruction lies elsewhere.

The objective of the CGJteamLab project was not to discover a new proof of Proposition I.1, but to investigate whether the theory developed in Book Zero is sufficiently rich to serve as an intermediate language for synthetic proofs.

The experiment shows that, once the existence problem for an equidistant point has been isolated, the entire argument can be expressed almost exclusively in terms of the theorems previously established in Book Zero.

From the perspective of the architecture of the theory, this is more significant than Proposition I.1 itself. It demonstrates that Book Zero is not merely a collection of auxiliary lemmas, but rather forms a new layer of abstraction between Hilbert's axioms and classical synthetic reasoning.

Schematically, the situation may be represented as

$$\text{Hilbert's axioms} \longrightarrow \text{Book Zero} \longrightarrow \text{Proposition I.1.}$$

In this sense, Proposition I.1 constitutes the first experimental test confirming that Book Zero can serve as a language of synthetic proofs for the further reconstruction of Euclidean geometry.

7 Relationship with the Foundations

Proposition I.1 has no earlier proposition of Book I on which it can depend. It therefore exposes the foundational boundary of the reconstruction more directly than any later proposition.

Euclid begins Book I with a circle-based construction: two circles with equal radii are drawn from the endpoints of the given segment, and an intersection point supplies the third vertex of the equilateral triangle.

In the Hilbert reconstruction, however, the essential issue is not segment congruence. Constructing points at prescribed distances is already supported by the congruence layer. The difficult step is to obtain a point realizing both distance conditions simultaneously and in the required nondegenerate configuration.

Thus I.1 immediately tests the interaction of

$$\text{incidence} \quad + \quad \text{order} \quad + \quad \text{congruence}$$

at the very beginning of Book I.

8 Hilbert and Book Zero Components Used

8.1 Segment Congruence

The equilateral conclusion is expressed synthetically:

$$AB \cong AC \qquad \text{and} \qquad AB \cong BC.$$

No numerical length function is used.

8.2 Betweenness Case Analysis

The reconstruction separates the possible collinear orders of three points. The three cases

$$A - B - C, \qquad B - A - C, \qquad A - C - B$$

are handled explicitly.

This is one of the earliest examples of a recurring formalization phenomenon: a configuration that is visually excluded in Euclid's diagram must be ruled out logically from order and congruence.

8.3 Book Zero Order Lemmas

The contradiction in each collinear branch depends on elementary results expressing that a proper part of a segment cannot be congruent to the whole segment.

These low-level theorems are precisely the kind of facts collected in Book Zero so that Book I proofs do not repeatedly reconstruct elementary betweenness arithmetic.

8.4 Noncollinearity

An equilateral triangle must be a genuine triangle. The formal construction therefore requires not only three pairwise congruent side relations but also a proof that the third point does not lie on the given line.

This nondegeneracy obligation is explicit in Lean and becomes an important pattern throughout the rest of Book I.

9 Formalization Notes

The complete implementation is available in the repository. Two excerpts show the mathematical structure.

Existence and the two equal-distance facts arrive together:

```
obtain <C, hAC, hBC> :=
  hilbert_equidistant_point_exists Geo A B hAB
```

After degeneracies are excluded, collinearity is split into the three betweenness cases:

```
rcases
    hilbert_between_trichotomy
      Geo A B C hAB hBCne hACne hCol with
  hABC | hBAC | hACB
```

Each branch is closed by `bookZero_45_partNotEqualWhole`, with the necessary congruence orientation supplied by symmetry or endpoint reversal. Those branch details are verification bookkeeping; the common geometric idea is simply that a proper part cannot equal the whole.

10 Dependency Summary

```
Classical:
Postulate III + circle intersection + Common Notion 1 -> I.1

Formal:
hilbert_equidistant_point_exists
        +
null-segment exclusion
        +
betweenness trichotomy
        +
bookZero_45_partNotEqualWhole
        |
        v
euclid_proposition_1
```

The formal theorem therefore replaces Euclid's circle-intersection mechanism by an abstract equidistant-point existence theorem and an explicit proof of noncollinearity.

11 Conclusion

Proposition I.1 is the natural starting point of Euclid's Book I, but its formal reconstruction immediately reveals that "starting point" does not mean "foundationally primitive."

Euclid's two-circle diagram suppresses elementary facts about point order, segment comparison, and nondegeneracy. Once these are made explicit, the proof reaches below Book I into the Hilbert and Book Zero layers.

The reconstruction therefore establishes an important theme for the entire project at the very first proposition: Book I is the visible classical development, while Book Zero supplies the elementary synthetic language needed to make its diagrammatic reasoning precise.

From this point onward, later propositions can reuse that language instead of rebuilding it. Proposition I.1 is consequently both the first Euclidean construction and the first demonstration of why the formal reconstruction requires a substantial interface beneath Euclid's text.